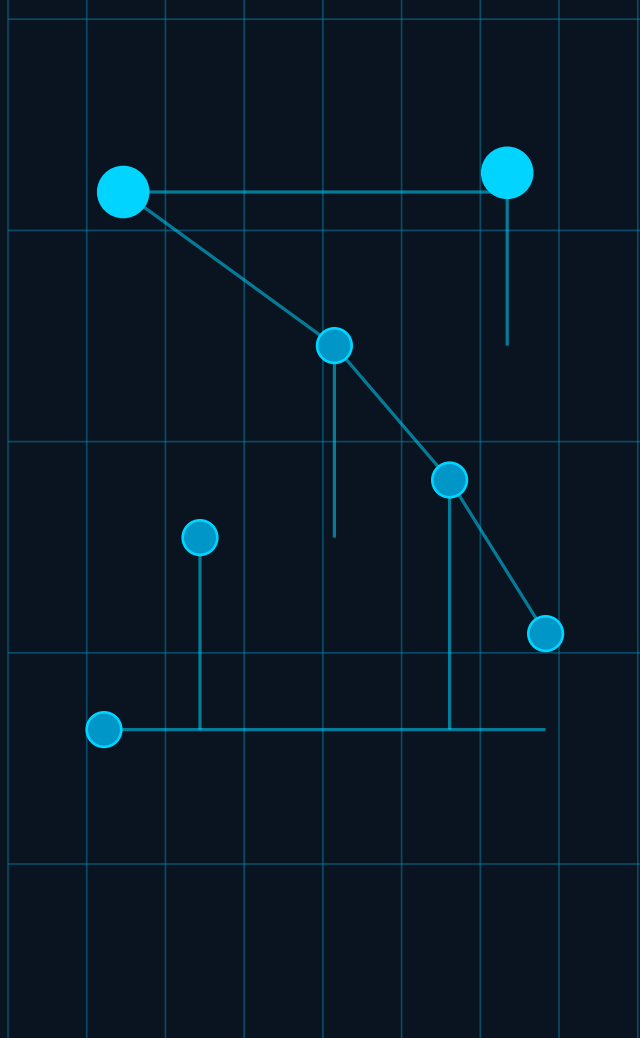


Improving Risk Analysis & Defense Prioritization

Modified PageRank Ranking · 0/1 Knapsack Optimization
Budget-Constrained Defense Selection

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The Cybersecurity Challenge

- Modern systems are composed of multiple interconnected components, massively expanding the attack surface.
- Attackers exploit vulnerabilities in sequence → complex multi-stage attacks that are hard to detect.
- Attack graphs model all possible paths an attacker can take through a system.
- Nodes = system states or vulnerabilities; Edges = attacker transitions between states.
- Securing every vulnerability is impractical due to limited budget, time, and manpower.
- Traditional methods treat vulnerabilities independently — no system-wide context.

→ *Need for intelligent, data-driven defense prioritization*

60%

of breaches exploit known vulnerabilities

~\$4.9M

average cost of a data breach (2024)

200+

days avg. attacker dwell time in a network

PageRank Algorithm

Graph Ranking

Evaluates node importance by graph connectivity. Widely adapted for network-level vulnerability prioritization.

DefenceRank

Attack Graphs

Extension of PageRank for attack graphs. Ranks nodes by structural importance to prioritize defensive actions.

Time-To-Compromise (TTC)

Risk Metrics

Estimates time required for an attacker to exploit a vulnerability. Used as a risk proxy in several frameworks.

Knapsack Optimization

Optimization

0/1 knapsack problem applied to defense resource allocation under strict budget constraints.

 **Common Limitations:** Ignore failure probability · No attacker behavior modeling · Poor budget integration · Static models

Gap 1

Failure Probability Ignored

Existing models assume attacks always succeed. Real-world vulnerability exploitation has a quantifiable failure rate that significantly affects risk assessment.

Gap 2

No Attacker Behavior Modeling

Most approaches ignore path preference — the tendency of attackers to reuse familiar, high-success attack paths — leading to inaccurate risk rankings.

Gap 3

Static & Non-Adaptive Models

Current methods provide point-in-time analysis and do not adapt to evolving attack strategies or dynamically changing network configurations.

Gap 4

Limited Budget Integration

Very few frameworks incorporate real-world resource constraints. Without budget modeling, recommended defenses are often impractical to deploy.

Objective

Design and develop a model that improves defense prioritization by handling complex attack graphs, probabilistic uncertainty, attacker behavior, and budget constraints to maximize security impact within resource limits.



Complex Graph Structures

Handle multi-path, cyclic, and hierarchical attack graph topologies with cross-links and back edges.



Probabilistic Uncertainty

Incorporate failure probability (fp) to reflect realistic attack success rates rather than assuming perfect exploitation.



Attacker Behavior (Path Preference)

Model the tendency of attackers to reuse frequent, high-success paths using the path preference (pp) attribute.



Budget-Constrained Optimization

Select the optimal subset of defenses that maximizes total security gain while respecting a strict cost budget B .

Proposed System

Four-Factor Risk Model

V — Node Value

Asset importance (5–10 scale)

pp — Path Preference

Attack path frequency / likelihood

fp — Failure Prob.

Probability of exploit failure

W — Graph Structure

Connectivity transition matrix

Parameters: $\alpha = 0.5$ · $\beta = 0.6$ · $\gamma = 0.7$

Mathematical Formulation

Main Rank Equation

$$R = \alpha \cdot T + (1 - \alpha) \cdot S$$

Intrinsic Importance

$$T = \beta \cdot pp + (1 - \beta) \cdot V$$

Propagated Importance

$$S = \gamma \cdot WR + (1 - \gamma) \cdot Z$$

Reliability Score

$$Z = -\log(fp)$$

Value Function

$$Val_i = R_i \times V_i$$

Knapsack Objective

$$\max \sum Val_i \cdot x_i$$

1

Load node attributes (V , pp , fp), graph edges, parameters α , β , γ , and defense costs

2

Compute intrinsic importance $T = \beta \cdot pp + (1 - \beta) \cdot V$ and normalize to unit sum

3

Compute reliability $Z = -\log(fp + \epsilon)$; normalize Z

4

Build transition matrix W from graph: uniform distribution over outgoing edges; dangling nodes $\rightarrow$ uniform

5

Initialize rank vector R uniformly: $R = [1/n, \dots, 1/n]$

6

Iteratively update: $WR = W^T R$; $S = \gamma \cdot WR + (1 - \gamma) \cdot Z$; $R_{\text{new}} = \alpha \cdot T + (1 - \alpha) \cdot S$; normalize

7

Convergence check: $\|R_{\text{new}} - R\| < 1 \times 10^{-6}$ (max 100 iterations)

8

Compute value function: $\text{Val}_i = R_i \times V_i$ for each node i

9

Run 0/1 knapsack DP: $dp[i][w] = \max(dp[i-1][w], \text{Val}_i + dp[i-1][w - C_i])$

10

Traceback DP table to recover selected defense nodes

Python

Pandas

NumPy

CSV Input

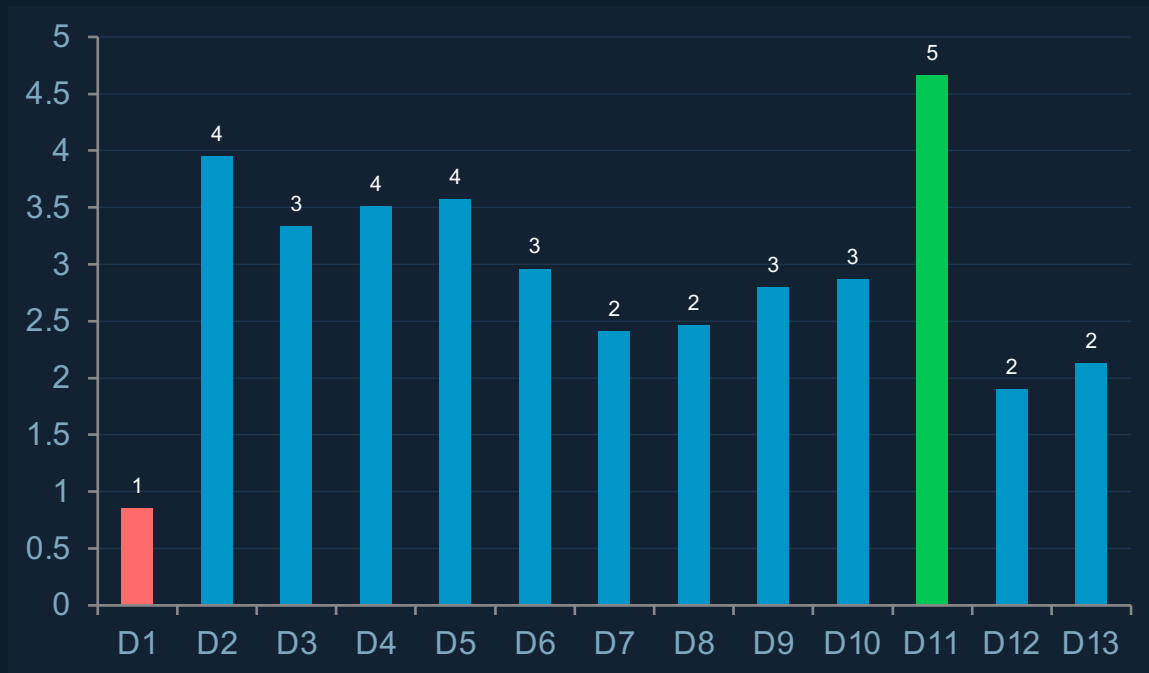
4 Files

 $\alpha=0.5$ $\beta=0.6$ $\gamma=0.7$

ID	Graph Topology	n	Description
D1	Main Attack Graph	30	Complex forward + cross-links
D2	Linear Chain	6	Sequential single-path route
D3	Binary Tree	7	Hierarchical branching structure
D4	Star Topology	7	Centralized HUB + leaf nodes
D5	Parallel Paths	8	Two independent paths → target
D6	Cycle + Cross-Links	8	Circular paths + shortcut edges
D7	Sparse Random	10	Low-density random connectivity
D8	Layered 3x3 Tiers	9	Structured tiered network
D9	Dual-Hub & Spoke	9	Two hubs controlling spoke nodes
D10	Dense Mesh	8	High-density interconnections
D11	Fully Connected	5	Every node connected to all others
D12	Hierarchical Tree	15	Multi-level parent-child structure
D13	Mixed + Back Edges	10	Forward paths with feedback loops

Results — Defense Selection

Security Gain per Dataset (Budget = 7)



D11 (Fully Connected, $n=5$) achieves highest gain — high-value nodes within budget reach.
D1 ($n=30$) lowest — large pool limits coverage at fixed budget.

Top Security Gains

D11 Fully Connected

Gain: 4.6639

D2 Linear Chain

Gain: 3.9529

D5 Parallel Paths

Gain: 3.5725

D4 Star Topology

Gain: 3.5087

Results — Optimal Defense Sets

DS	Topology	n	Selected Nodes	Cost	Sec. Gain	Top Node
D1	Main Attack Graph	30	E, Y, AB	7	0.8515	AD
D2	Linear Chain	6	N5, N6	7	3.9529	N6
D3	Binary Tree	7	R, L1, L3	7	3.3324	R3
D4	Star Topology	7	S1, S3, S4, S5	7	3.5087	HUB
D5	Parallel Paths	8	D1, E	7	3.5725	E
D6	Cycle + Cross-Links	8	P6, P7, P8	7	2.9595	P6
D7	Sparse Random	10	V4, V6	7	2.4134	V6
D8	Layered 3x3 Tiers	9	T3B, T3C	7	2.4617	T3C
D9	Dual-Hub & Spoke	9	S1, S3, S4, S5, S7	7	2.8008	H2
D10	Dense Mesh	8	M2, M6, M7	7	2.8732	M8
D11	Fully Connected	5	F1, F5	7	4.6639	F5
D12	Hierarchical Tree	15	F, I, K, L, N	7	1.8992	L
D13	Mixed + Back Edges	10	X8, X9	7	2.1344	X9

Results — DP vs Brute Force Validation

12 / 12

Datasets with
Perfect Match (100%)

100%

Accuracy on
all feasible datasets

1

Dataset skipped
($n=30 \rightarrow 2^{30}$ infeasible)

DS	n	DP Value	BF Value	Match	Accuracy
D1	30	0.8515	N/A ($n>20$)	SKIPPED	—
D2	6	3.9529	3.9529	YES	100%
D3	7	3.3324	3.3324	YES	100%
D4	7	3.5087	3.5087	YES	100%
D5	8	3.5725	3.5725	YES	100%
D6	8	2.9595	2.9595	YES	100%
D7	10	2.4134	2.4134	YES	100%
D8	9	2.4617	2.4617	YES	100%
D9	9	2.8008	2.8008	YES	100%
D10	8	2.8732	2.8732	YES	100%
D11	5	4.6639	4.6639	YES	100%
D12	15	1.8992	1.8992	YES	100%
D13	10	2.1344	2.1344	YES	100%

*DP Knapsack matches globally
optimal brute-force solution
across ALL 12 feasible datasets
— implementation correctness
confirmed.*

Conclusion



Modified PageRank integrates node value, path preference, failure probability, and graph structure into a unified risk model



0/1 Knapsack DP achieves 100% accuracy vs. brute force across all 12 feasible datasets ($n \leq 20$)



Evaluated across 13 diverse topologies — linear, tree, star, cycle, mesh, hierarchical, and mixed graphs (5–30 nodes)



Integrating fp and pp produces more realistic rankings than topology-only approaches (DefenceRank, TTC)

A practical, budget-aware cybersecurity framework that balances theoretical rigor with real-world applicability.

Future Work

- Dynamic attack graph modeling for real-time adaptation
- Validation on NVD / CVE real-world datasets
- Integer Linear Programming for very large graphs
- Multi-objective optimization balancing security gain & operational impact